

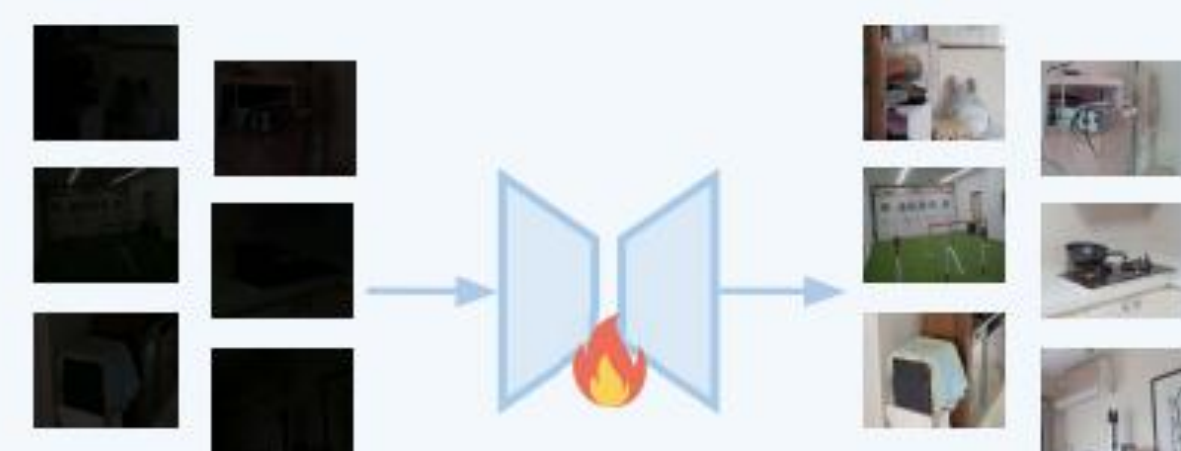


Problem & Motivation

Problem

- Existing methods, whether supervised, unsupervised, or zero-shot, rely on **auxiliary loss functions** and **empirically selected hyperparameters**.
- They yield strong results on the datasets used for **evaluation** but often exhibit **limited generalization**.

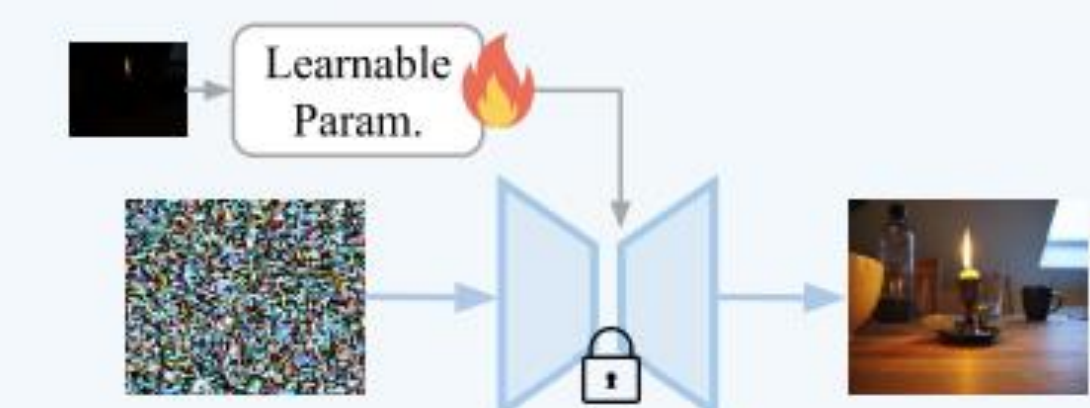
LLIE Method Taxonomy



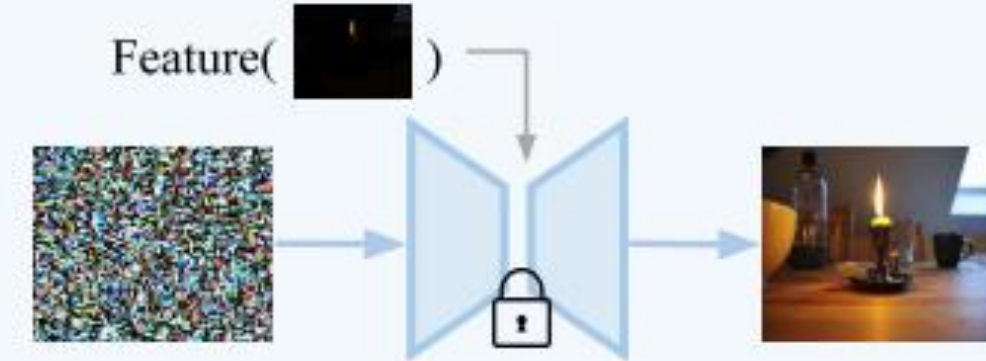
(A) Supervised & Unsupervised



(B) Instance-wise Optimization (w/o Pretrained Model)



(C) Instance-wise Optimization (w/ Pretrained Model)

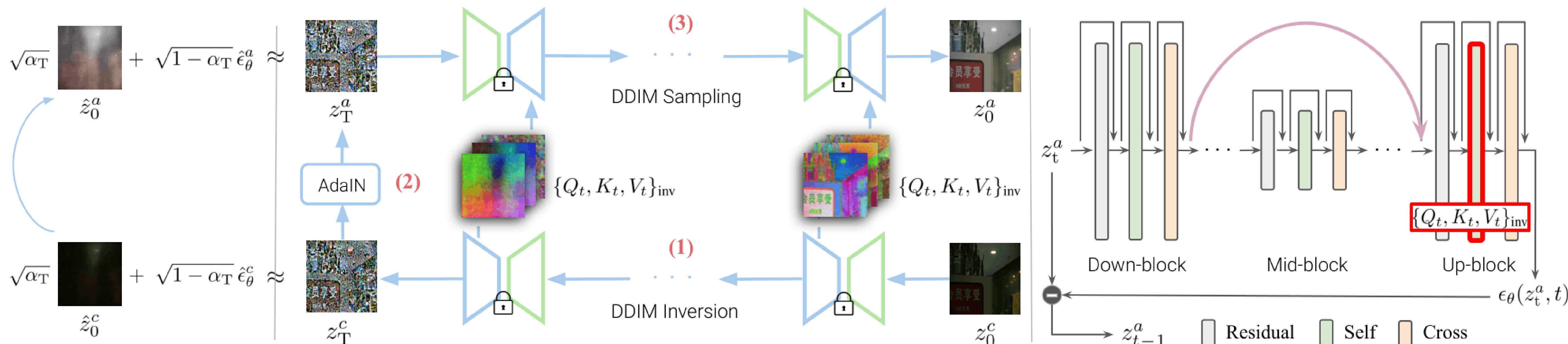


(D) Optimization-free (Ours)

Contribution

- To our knowledge, MR. Illuminate is the **first** zero-shot LLIE method that does not rely on auxiliary objectives and empirically selected hyperparameters.
- To our knowledge, MR. Illuminate is also the **first** zero-shot AWB method.
- Our method demonstrates **reconstruction consistency** across inputs of the same scene captured under different lighting and noise conditions

How Does It Work?



MR: Modulate & Refine

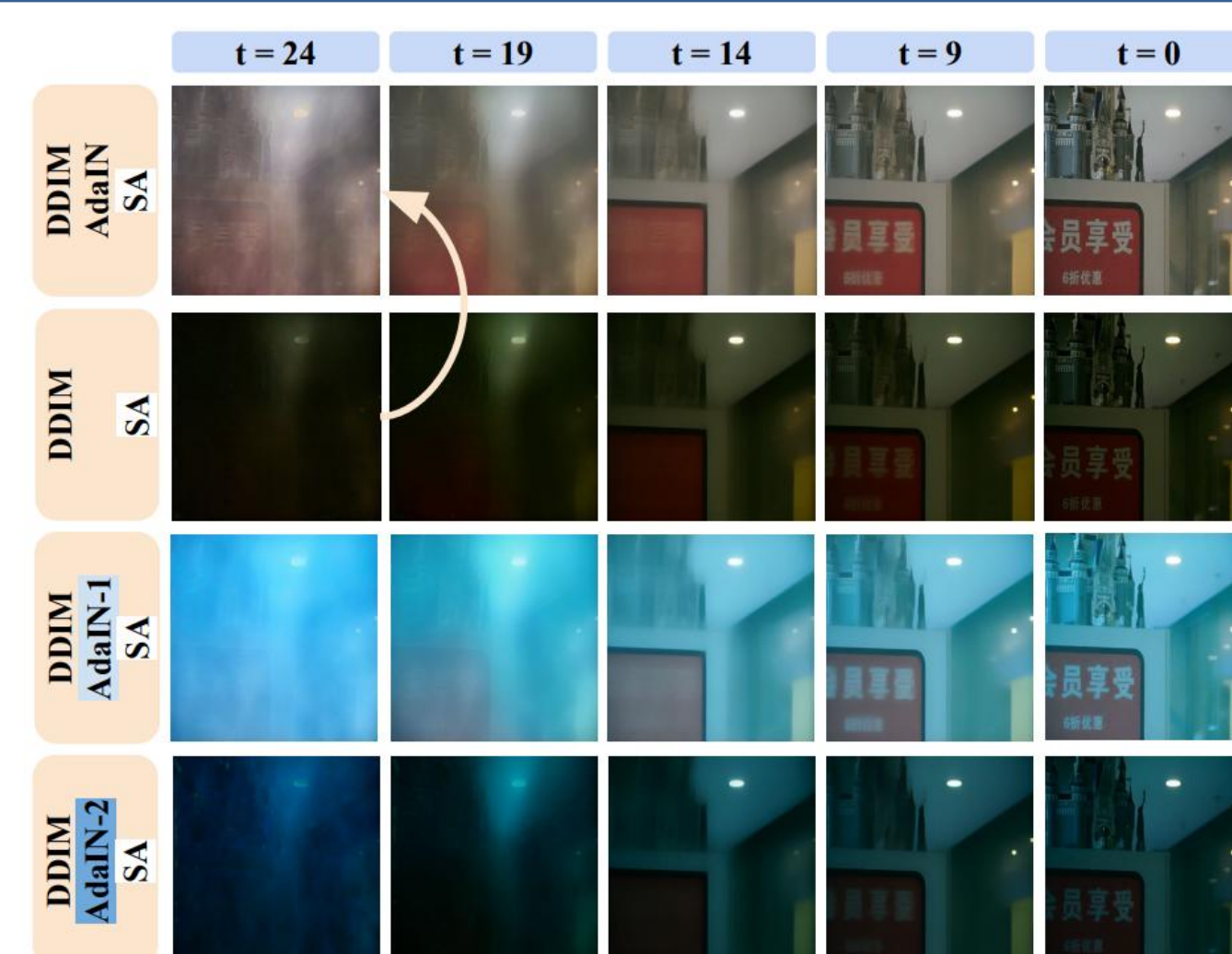


Modulate

AdaIN “modulates” the initial noise to produce a well-illuminated output while preserving input structure.

Refine

We gradually “refine” the intermediate latent representation through self-attention injection at each denoising step.



Results

